Classical provability of higher-order infinitary geometric implications implies intuitionistic provability

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Allowed standard prerequisites (editorial).

- Ordinary higher-order natural deduction and standard continuation monad unit/bind identities. Original reference38 FSCD2025.34 explicitly cited; complete relevant author Sections2–6 including Proposition2 monad proofs and Lemma16 finite translation cases are included as support, while current countably-infinitary additions are retained from FSCD2026.33.

Higher-order infinitary syntax and deduction

Original definition (1). *Types and terms are defined inductively:*

$$\begin{array}{rcl} \tau, \sigma & ::= & \iota \mid o \mid o^I \mid \tau \to \sigma, \\ t, u, A_i, F & ::= & x \mid c \mid t\,u \mid \lambda x.t \mid \{i \mapsto A_i\}_I \mid \pi_j F. \end{array}$$

Here ι is the type of individuals, o is the type of propositions, o^I is the type of families of propositions indexed by a countable set I, x is a variable, c is a constant, $\{i \mapsto A_i\}_I$ is the family of propositions $(A_i)_{i\in I}$, and $\pi_j F$ is the projection on the j-th component of F.

The typing rules are the usual ones, extended with: if for every $i \in I$ we have A_i of type o then $\{i \mapsto A_i\}_I$ is of type o^I; if F is of type o^I then $\pi_j F$ is of type o. We write Λ_τ for the set of terms of type τ. We restrict terms so that they do not have infinitely many free variables. We write $t[x \leftarrow u]$ for the term t in which x has been substituted by u.

Original definition (3). *We consider the following rules:*

$$(\lambda x.t)u \hookrightarrow t[x \leftarrow u], \quad \pi_j\{i \mapsto A_i\}_I \hookrightarrow A_j, \quad \lambda x.tx \hookrightarrow t \ (x \notin \mathrm{fv}(t)), \quad \{i \mapsto \pi_i F\}_I \hookrightarrow F.$$

The reduction relation $\hookrightarrow$ is the smallest relation that contains these rules and such that:

- *if $t \hookrightarrow t'$, then $tu \hookrightarrow t'u$, $ut \hookrightarrow ut'$ and $\lambda x.t \hookrightarrow \lambda x.t'$;*
- *if $A_i \hookrightarrow B_i$ for every $i \in J$, where $J \neq \varnothing$ and $J \subseteq I$, and $A_i = B_i$ otherwise, then $\{i \mapsto A_i\}_I \hookrightarrow \{i \mapsto B_i\}_I$;*
- *if $F \hookrightarrow F'$ then $\pi_j F \hookrightarrow \pi_j F'$.*

The relation $\equiv$ is the reflexive, symmetric and transitive closure of $\hookrightarrow$. The $\beta\pi$-reduction $\hookrightarrow_{\beta\pi}$ and $\beta\pi$-conversion $\equiv_{\beta\pi}$ are defined like $\hookrightarrow$ and $\equiv$, but only consider the β-reduction rule and the projection rules.

There are particular constants, called logical constants, for defining the connectives and quantifiers: contradiction $\bot$ of type o, implication $\Rightarrow$ of type $o \to o \to o$, quantifiers $\forall_\tau$ and $\exists_\tau$ of type $(\tau \to o) \to o$, infinite conjunction $\bigwedge_I$ and infinite disjunction $\bigvee_I$ of type $o^I \to o$. Terms $\forall_\tau(\lambda x.A)$ and $\exists_\tau(\lambda x.A)$ are abbreviated as $\forall x.A$ and $\exists x.A$. Negation is $\neg A := A \Rightarrow \bot$ and $A \Leftrightarrow B := (A \Rightarrow B) \wedge (B \Rightarrow A)$. Formulas are terms of type o. A context Γ is a finite sequence of formulas. We write $\Gamma \vdash A$ when A is derivable from Γ. We write $\mathrm{fv}(t_1, \dots, t_n)$ for the set of free variables that occur in $t_1, \dots, t_n$. The ordinary natural-deduction rules are obtained by ignoring the cut-free/neutral adornments of Figure1 in the original paper. The finite higher-order natural-deduction rules, conversions and monad facts are retained in the complete collected author precursor Sections2–6 below. The additional infinitary rules, with those adornments erased as specified by the authors, are:

$$\frac{\Gamma \vdash \pi_i F \text{ for every } i \in I}{\Gamma \vdash \bigwedge_I F}\ \text{AndInf– I} \qquad \frac{\Gamma \vdash \bigwedge_I F}{\Gamma \vdash \pi_j F}\ \text{AndInf– E}$$

$$\frac{\Gamma \vdash \pi_j F}{\Gamma \vdash \bigvee_I F}\ \text{OrInf– I} \qquad \frac{\Gamma \vdash \bigvee_I F \quad \Gamma, \pi_i F \vdash C \text{ for every } i \in I}{\Gamma \vdash C}\ \text{OrInf– E.}$$

There are versions of AndInf-E and OrInf-I for every $j \in I$. Derivation trees possibly have an infinite width, but each branch has a finite height. The formula $A \wedge B$ (respectively $A \vee B$) is defined by $\bigwedge_I \{i \mapsto A_i\}_{\{0,1\}}$ (respectively $\bigvee_I \{i \mapsto A_i\}_{\{0,1\}}$), where $A_0 = A$ and $A_1 = B$. The finite conjunction and disjunction rules are particular cases of those for the infinite conjunction and disjunction. We consider either classical logic CL or intuitionistic logic IL – in which we do not have the PEM rule. Weakening, contraction and exchange are admissible.

Barr's theorem for higher-order geometric theories

Geometric formulas are built using finite conjunctions, infinite disjunctions and existential quantifiers. Geometric implications are formulas of the form $\forall \vec{x}.(A \Rightarrow B)$, where A and B are geometric formulas and $\vec{x}$ is a finite sequence of variables (possibly empty). A theory $\mathcal{T}$ is geometric when all its axioms are geometric implications. Throughout this section, we consider a higher-order infinitary logic with infinite disjunctions, but with finite conjunctions.

Original definition (39, Friedman's translation)**.** *Let A be a formula in higher-order infinitary logic. Its Friedman's translation is $A^T := TA_T$, where $t \mapsto t_T$ is inductively defined by:*

$$\begin{array}{rcl} x_T & := & x, \\ c_T & := & \begin{cases} \lambda p.\forall x.T(px) & \text{if } c = \forall, \\ \lambda p.\lambda q.p \Rightarrow Tq & \text{if } c = \Rightarrow, \\ c & \text{otherwise,} \end{cases} \\ (\lambda x.t)_T & := & \lambda x.t_T, \\ (tu)_T & := & t_T u_T, \\ (\{i \mapsto A_i\}_I)_T & := & \{i \mapsto (A_i)_T\}_I, \\ (\pi_j F)_T & := & \pi_j F_T. \end{array}$$

As we are in higher-order logic, our translation must commute with substitution and preserve convertibility. Note that all the results of this section also work if we only consider the β-reduction rule and the projection rules.

Original proposition (40)**.** *Let t, u be terms and A, B be higher-order infinitary formulas.*

1. $(t[z \leftarrow w])_T = t_T[z \leftarrow w_T]$.
2. $(A[z \leftarrow w])^T = A^T[z \leftarrow w_T]$.
3. *If $t \equiv u$ then $t_T \equiv u_T$.*
4. *If $A \equiv B$ then $A^T \equiv B^T$.*

Proof. The first item is proved by induction on the term t. The second item is a corollary of the first item. The third item is proved by induction on $t \equiv u$. The fourth item is a corollary of the third item. □

Let us remark that the operator $TA := (A \Rightarrow R) \Rightarrow R$ corresponds to a monad operator in the sense of original reference38, meaning that it satisfies the (unit) and (bind) properties:

$$\vdash_L A \Rightarrow TA, \tag{unit}$$

$$\vdash_L (A \Rightarrow TB) \Rightarrow (TA \Rightarrow TB). \tag{bind}$$

It follows that the following properties hold [38].

Original proposition (41)**.** *Let A, B be formulas and P be a predicate.*

1. $\vdash_{\mathsf{IL}} TTA \Leftrightarrow TA$.
2. $\vdash_{\mathsf{IL}} (A \Rightarrow B) \Rightarrow (TA \Rightarrow TB)$.
3. $\vdash_{\mathsf{IL}} T(A \wedge B) \Leftrightarrow (TA \wedge TB)$.
4. $\vdash_{\mathsf{IL}} T(A \Rightarrow B) \Rightarrow (TA \Rightarrow TB)$.
5. $\vdash_{\mathsf{IL}} T(A \Rightarrow TB) \Leftrightarrow (TA \Rightarrow TB)$.
6. $\vdash_{\mathsf{IL}} T(\forall x.T(Px)) \Leftrightarrow \forall x.T(Px)$.
7. $\vdash_{\mathsf{IL}} T(\exists x.T(Px)) \Leftrightarrow T(\exists P)$.

Original theorem (42)**.** *Let $TA = (A \Rightarrow R) \Rightarrow R$ with parameter R. Let Γ, A be higher-order infinitary formulas. Let $\mathcal{T}$ be a higher-order geometric theory. If $\Gamma \vdash_{\mathsf{CL}} A$ in $\mathcal{T}$ then $\Gamma^T \vdash_{\mathsf{IL}} A^T$ in $\mathcal{T}$.*

Proof. We proceed by induction on the derivation. Most of the cases are direct consequences of Proposition41 and are detailed in [38, see Lemma16]. We only show the important cases.

OrInf-I: By induction hypothesis, we have $\Gamma^T \vdash_{\mathsf{IL}} T(\pi_j F)_T$. Using Ax, we have $\Gamma^T, (\pi_j F)_T \vdash_{\mathsf{IL}} (\pi_j F)_T$. Using OrInf-I, we get $\Gamma^T, (\pi_j F)_T \vdash_{\mathsf{IL}} \bigvee_I F$. Using Imp-I and Proposition41(2), we derive $\Gamma^T \vdash_{\mathsf{IL}} T(\pi_j F)_T \Rightarrow T(\bigvee_I F)$. We conclude using the induction hypothesis.

OrInf-E: By induction hypotheses, we have $\Gamma^T \vdash_{\mathsf{IL}} T(\bigvee_I F)$ and $\Gamma^T, (\pi_i F)_T \vdash_{\mathsf{IL}} TC_T$ for every $i \in I$. By Ax, we have $\Gamma^T, \bigvee_I F \vdash_{\mathsf{IL}} \bigvee_I F$. By weakening of the induction hypotheses, we have $\Gamma^T, \bigvee_I F, (\pi_i F)_T \vdash_{\mathsf{IL}} TC_T$ for every $i \in I$. Therefore, we derive $\Gamma^T, \bigvee_I F \vdash_{\mathsf{IL}} TC_T$ using OrInf-E. We get $\Gamma^T \vdash_{\mathsf{IL}} T(\bigvee_I F) \Rightarrow TC_T$ using Imp-I and (bind). We conclude using Imp-E with the first induction hypothesis.

PEM: We prove $\Gamma^T \vdash_{\mathsf{IL}} T(A \vee \neg A)_T$ using the following author derivation (the rows retain the original successive inference labels):

$$\begin{array}{rll}
\text{Ax}: & \Gamma^T, B \Rightarrow R, A_T, \bot \Rightarrow R & \vdash_{\mathsf{IL}} A_T \\
\text{Or– IL}: & \Gamma^T, B \Rightarrow R, A_T, \bot \Rightarrow R & \vdash_{\mathsf{IL}} B \\
\text{Imp– E with } B \Rightarrow R: & \Gamma^T, B \Rightarrow R, A_T, \bot \Rightarrow R & \vdash_{\mathsf{IL}} R \\
\text{Imp– I} \times 2: & \Gamma^T, B \Rightarrow R & \vdash_{\mathsf{IL}} A_T \Rightarrow ((\bot \Rightarrow R) \Rightarrow R) \\
\text{Or– IR}: & \Gamma^T, B \Rightarrow R & \vdash_{\mathsf{IL}} B \\
\text{Imp– E with } B \Rightarrow R: & \Gamma^T, B \Rightarrow R & \vdash_{\mathsf{IL}} R \\
\text{Imp– I}: & \Gamma^T & \vdash_{\mathsf{IL}} ((A \vee (A \Rightarrow ((\bot \Rightarrow R) \Rightarrow R))) \Rightarrow R) \Rightarrow R
\end{array}$$

where B is an abbreviation for the formula $A_T \vee (A_T \Rightarrow ((\bot \Rightarrow R) \Rightarrow R))$.

□

Original Remark43. The And-I case crucially depends on $\vdash_{\mathsf{IL}} (TA \wedge TB) \Rightarrow T(A \wedge B)$. The proof of this property does not extend to the infinite case. This is why, in this section, we consider finite conjunctions and not infinite conjunctions. Geometric formulas do not contain implications or universal quantifiers, while the translation $t \mapsto t_T$ only inserts T after universal quantifiers and implications. It follows that for any geometric formula A, we directly have $A_T = A$. The same result does not hold for geometric implications $\forall \vec{x}.(A \Rightarrow B)$. But geometric implications imply their translation.

Original lemma (44)**.** *Let A, B be higher-order geometric formulas, and $\vec{x}$ be a finite sequence of variables (possibly empty). We have $\vdash_{\mathsf{IL}} (\forall \vec{x}.(A \Rightarrow B)) \Rightarrow (\forall \vec{x}.(A \Rightarrow B))_T$.*

Proof. Since $(A \Rightarrow B)_T = A \Rightarrow TB$, we directly have $\vdash_{\mathsf{IL}} (A \Rightarrow B) \Rightarrow (A \Rightarrow B)_T$ using (unit) on B. If $\vec{x}$ is empty, this achieves the case. Suppose that $\vec{x} = x_1, \ldots, x_n$. We derive $\vdash_{\mathsf{IL}} (A \Rightarrow B) \Rightarrow T(A \Rightarrow B)_T$ using (unit) on $(A \Rightarrow B)_T$. Using All-I, we derive $\vdash_{\mathsf{IL}} \forall x_n.((A \Rightarrow B) \Rightarrow T(A \Rightarrow B)_T)$. Then we use the distributivity of $\forall$ over $\Rightarrow$ to derive $\vdash_{\mathsf{IL}} (\forall x_n.(A \Rightarrow B)) \Rightarrow (\forall x_n.T(A \Rightarrow B)_T)$, that is $\vdash_{\mathsf{IL}} (\forall x_n.(A \Rightarrow B)) \Rightarrow (\forall x_n.(A \Rightarrow B))_T$. We iterate this procedure to get $\vdash_{\mathsf{IL}} (\forall \vec{x}.(A \Rightarrow B)) \Rightarrow (\forall \vec{x}.(A \Rightarrow B))_T$. □

Original theorem (45)**.** *Let $\mathcal{T}$ be a higher-order geometric theory, A and B be higher-order geometric formulas, and $\vec{x}$ be a finite sequence of variables (possibly empty). If $\vdash_{\mathsf{CL}} \forall \vec{x}.(A \Rightarrow B)$ in $\mathcal{T}$ then $\vdash_{\mathsf{IL}} \forall \vec{x}.(A \Rightarrow B)$ in $\mathcal{T}$.*

Proof. Suppose that $\vdash_{\mathsf{CL}} \forall \vec{x}.(A \Rightarrow B)$ in $\mathcal{T}$. Then there exists some geometric implications Θ of $\mathcal{T}$ such that $\Theta \vdash_{\mathsf{CL}} \forall \vec{x}.(A \Rightarrow B)$. Using All-E, we instantiate the bound variables $\vec{x}$ by fresh variables also named $\vec{x}$ (without loss of generality) and obtain $\Theta \vdash_{\mathsf{CL}} A \Rightarrow B$. Using Weak and Imp-E, we get $\Theta, A \vdash_{\mathsf{CL}} B$. Applying Theorem42, we derive $\Theta^T, A^T \vdash_{\mathsf{IL}} B^T$, that is $\Theta^T, A \vdash_{\mathsf{IL}} (B \Rightarrow R) \Rightarrow R$. By Lemma44, we derive $\Theta, A \vdash_{\mathsf{IL}} (B \Rightarrow R) \Rightarrow R$. Choosing $R := B$, we get $\Theta, A \vdash_{\mathsf{IL}} B$. Using Imp-I and All-I, we derive $\Theta \vdash_{\mathsf{IL}} \forall \vec{x}.(A \Rightarrow B)$. We conclude that $\vdash_{\mathsf{IL}} \forall \vec{x}.(A \Rightarrow B)$ in $\mathcal{T}$. □

Exact author reference for finite translation cases

[38] Thomas Traversié. Monad Translations for Higher-Order Logic. FSCD2025, LIPIcs337, Article34, pp34:1–14. DOI10.4230/LIPIcs.FSCD.2025.34. The complete directly cited Lemma16, its proof, and all connected author Sections2–6 (definitions, finite rules, monad embedding and constructive translation proofs), together with the original bibliography, are retained below as support only; they are not additional counted targets.

2 Preliminaries

The syntax and inference rules of higher-order logic are recalled in Section 2.1. The basic notions about monad operators are summarized in Section 2.2.

2.1 Higher-Order Logic

Higher-order logic is modeled using Church's simple type theory [4]. Types are defined inductively: ι is the type of individuals, o is the type of propositions, and if τ and σ are types then $\tau \to \sigma$ is a type. For every type τ, let $\mathcal{V}_\tau$ be the set of variables of type τ and $\mathcal{C}_\tau$ be a set of constants of type τ. The set of variables $\mathcal{V} := \bigcup_\tau \mathcal{V}_\tau$ and the set of constants $\mathcal{C} := \bigcup_\tau \mathcal{C}_\tau$ are assumed to be disjoint. For any set of constants $\mathcal{C}$, the sets $\Lambda^{\mathcal{C}}_\tau$ of terms of type τ are defined by induction:

- For every $x \in \mathcal{V}_\tau$, $x \in \Lambda^{\mathcal{C}}_\tau$.
- For every $c \in \mathcal{C}_\tau$, $c \in \Lambda^{\mathcal{C}}_\tau$.
- For every $x \in \mathcal{V}_\tau$ and $t \in \Lambda^{\mathcal{C}}_\sigma$, then $(\lambda x.\ t) \in \Lambda^{\mathcal{C}}_{\tau\to\sigma}$.
- For every $t \in \Lambda^{\mathcal{C}}_{\tau\to\sigma}$ and $u \in \Lambda^{\mathcal{C}}_\tau$, then $(t\ u) \in \Lambda^{\mathcal{C}}_\sigma$.

$\lambda x.\ t$ is a λ-abstraction and $t\ u$ is an application. Computation is introduced in this λ-calculus thanks to the β-reduction rule $(\lambda x.\ t)\ u \hookrightarrow t[x \leftarrow u]$, where $t[x \leftarrow u]$ corresponds to the term t in which x has been substituted by u. We denote $\equiv_\beta$ the congruence generated by β-reduction. We write $\mathrm{FV}(t_1, \ldots, t_n)$ for the set of free variables that occur in the terms $t_1, \ldots, t_n$.

Formulas are terms of type o. There are particular constants defining the logical connectives and quantifiers: contradiction $\bot$ of type o, implication $\Rightarrow$, conjunction $\wedge$ and disjunction $\vee$ of type $o \to o \to o$, and quantifiers $\forall_\tau$ and $\exists_\tau$ of type $(\tau \to o) \to o$. For convenience, terms of the form $\forall_\tau\ (\lambda x.\ A)$ and $\exists_\tau\ (\lambda x.\ A)$ are simply abbreviated as $\forall x.A$ and $\exists x.A$. The negation is defined by $\neg A := A \Rightarrow \bot$ and the logical biconditional $\Leftrightarrow$ is defined by $A \Leftrightarrow B := (A \Rightarrow B) \wedge (B \Rightarrow A)$. Contexts Γ are finite sequences of formulas.

In the rest of this paper, we consider a logic L which is either minimal logic (ML), intuitionistic logic (IL), or classical logic (CL). The natural deduction rules for CL are given in Figure 1. In IL, we do not consider the principle of excluded middle PEM, and in ML we neither consider PEM nor the principle of explosion Bot-E. We write $\Gamma \vdash_{\mathsf{L}} A$ when $\Gamma \vdash A$ is derivable in the logic L.

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \Rightarrow B}\ \textsc{Imp-I} \qquad \frac{\Gamma \vdash A \Rightarrow B \qquad \Gamma \vdash A}{\Gamma \vdash B}\ \textsc{Imp-E}$$

$$\frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \wedge B}\ \textsc{And-I} \qquad \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash A}\ \textsc{And-EL} \qquad \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash B}\ \textsc{And-ER}$$

$$\frac{\Gamma \vdash A}{\Gamma \vdash A \vee B}\ \textsc{Or-IL} \qquad \frac{\Gamma \vdash B}{\Gamma \vdash A \vee B}\ \textsc{Or-IR} \qquad \frac{\Gamma \vdash A \vee B \qquad \Gamma, A \vdash C \qquad \Gamma, B \vdash C}{\Gamma \vdash C}\ \textsc{Or-E}$$

$$\frac{\Gamma \vdash A \qquad x \notin FV(\Gamma)}{\Gamma \vdash \forall x.A}\ \textsc{All-I} \qquad \frac{\Gamma \vdash \forall x.A}{\Gamma \vdash A[x \leftarrow t]}\ \textsc{All-E}$$

$$\frac{\Gamma \vdash A[x \leftarrow t]}{\Gamma \vdash \exists x.A}\ \textsc{Ex-I} \qquad \frac{\Gamma \vdash \exists x.A \qquad \Gamma, A \vdash C \qquad x \notin FV(\Gamma, C)}{\Gamma \vdash C}\ \textsc{Ex-E}$$

$$\frac{}{\Gamma, A, \Delta \vdash A}\ \textsc{Ax} \qquad \frac{\Gamma \vdash A \qquad A \equiv_\beta B}{\Gamma \vdash B}\ \textsc{Conv}$$

$$\frac{\Gamma \vdash \bot}{\Gamma \vdash A}\ \textsc{Bot-E} \qquad \frac{}{\Gamma \vdash A \vee \neg A}\ \text{PEM}$$

Figure 1: Natural deduction rules for classical logic.

We can also define, for every type τ, an equality symbol $=_\tau$ of type $\tau \to \tau \to o$. The symbols are infix, and we write $t = u$ when there is no ambiguity on the type τ. The natural deduction rules for equality are given in Figure 2. We write $\Gamma \vdash^*_{\mathsf{L}} A$ with $* \in \{\mathfrak{e}, \mathfrak{ep}, \mathfrak{ef}, \mathfrak{efp}\}$ when $\Gamma \vdash_{\mathsf{L}} A$ is derivable with additional inference rules: with Eq-I and Eq-E when $\mathfrak{e}$ occurs in $*$, with PropExt when $\mathfrak{p}$ occurs in $*$, and with FunExt when $\mathfrak{f}$ occurs in $*$.

$$\frac{}{\Gamma \vdash u = u}\ \textsc{Eq-I} \qquad \frac{\Gamma \vdash A[x \leftarrow u] \qquad \Gamma \vdash u = v}{\Gamma \vdash A[x \leftarrow v]}\ \textsc{Eq-E}$$

$$\frac{\Gamma \vdash f\ x = g\ x \qquad x \notin FV(\Gamma, f, g)}{\Gamma \vdash f = g}\ \textsc{FunExt} \qquad \frac{\Gamma \vdash A \Rightarrow B \qquad \Gamma \vdash B \Rightarrow A}{\Gamma \vdash A = B}\ \textsc{PropExt}$$

Figure 2: Natural deduction rules for equality.

2.2 Monad Operators

We say that T is a unary connective when it is a closed term of type $o \to o$. A unary connective T is a monad operator of L when the judgments (unit) and (bind) hold for any formulas A and B.

$\vdash_{\mathsf{L}} A \Rightarrow TA$ (unit)

$\vdash_{\mathsf{L}} (A \Rightarrow TB) \Rightarrow (TA \Rightarrow TB)$ (bind)

We use the terminology "monad operator" to avoid any confusion with monads in the sense of category theory. Monad operators are called lax modalities in [1] and nuclei in [22]. This definition of monad operators is equivalent to the axiomatization of strong monads in [6]. Following the Curry-Howard correspondence, monad operators coincide with the notion of monad in programming languages [15], for instance in Haskell.

Example 1.

There are many monad operators of minimal logic:

■

$TA := A \vee \bot$,

■

$T_R A := (A \Rightarrow R) \Rightarrow R$, with parameter R, which corresponds to the continuation monad,

■

$TA := \neg\neg A$, which is a special case of the previous one with $R := \bot$,

■

$T_R A := (A \Rightarrow R) \Rightarrow A$, with parameter R, which corresponds to the Peirce monad [6].

We recall some formulas about monad operators that will be used in the rest of the paper.

Proposition 2.

Let T be a monad operator of L, and A and B be formulas.

1. $\vdash_{\mathsf{L}} TTA \Leftrightarrow TA$
2. $\vdash_{\mathsf{L}} (A \Rightarrow B) \Rightarrow (TA \Rightarrow TB)$
3. $\vdash_{\mathsf{L}} T(A \wedge B) \Leftrightarrow (TA \wedge TB)$
4. $\vdash_{\mathsf{L}} T(A \Rightarrow B) \Rightarrow (TA \Rightarrow TB)$
5. $\vdash_{\mathsf{L}} T(A \Rightarrow TB) \Leftrightarrow (TA \Rightarrow TB)$
6. $\vdash_{\mathsf{L}} T(A \vee B) \Leftrightarrow T(TA \vee TB)$
7. $\vdash_{\mathsf{L}} T(\forall x.TA) \Leftrightarrow \forall x.TA$
8. $\vdash_{\mathsf{L}} T(\exists x.TA) \Leftrightarrow T(\exists x.A)$

Proof.

We only show the most interesting cases.

■

Item 3 ($\Rightarrow$): We have $\vdash_{\mathsf{L}} (A \wedge B) \Rightarrow A$ and $\vdash_{\mathsf{L}} (A \wedge B) \Rightarrow B$. Using Item 2 we easily derive $\vdash_{\mathsf{L}} T(A \wedge B) \Rightarrow (TA \wedge TB)$.

■

Item 3 ($\Leftarrow$): Using (bind), we directly have $\vdash_{\mathsf{L}} (A \Rightarrow T(A \wedge B)) \Rightarrow TA \Rightarrow T(A \wedge B)$ and $\vdash_{\mathsf{L}} (B \Rightarrow T(A \wedge B)) \Rightarrow TB \Rightarrow T(A \wedge B)$. Using these two facts and (unit), we easily derive $\vdash_{\mathsf{L}} TA \Rightarrow TB \Rightarrow T(A \wedge B)$. We conclude $\vdash_{\mathsf{L}} (TA \wedge TB) \Rightarrow T(A \wedge B)$.

■

Item 4: We know that $\vdash_{\mathsf{L}} ((A \Rightarrow B) \wedge A) \Rightarrow B$. We derive $\vdash_{\mathsf{L}} T((A \Rightarrow B) \wedge A) \Rightarrow TB$ using Item 2. By Item 3, we get $\vdash_{\mathsf{L}} (T(A \Rightarrow B) \wedge TA) \Rightarrow TB$, therefore we conclude $\vdash_{\mathsf{L}} T(A \Rightarrow B) \Rightarrow (TA \Rightarrow TB)$.

■

Item 5 ($\Rightarrow$): We have $\vdash_{\mathsf{L}} T(A \Rightarrow TB) \Rightarrow (TA \Rightarrow TTB)$ using Item 4. We conclude $\vdash_{\mathsf{L}} T(A \Rightarrow TB) \Rightarrow (TA \Rightarrow TB)$ using Item 1.

■

Item 5 ($\Leftarrow$): Using (unit), we derive $\vdash_{\mathsf{L}} (TA \Rightarrow TB) \Rightarrow (A \Rightarrow TB)$ and $\vdash_{\mathsf{L}} (A \Rightarrow TB) \Rightarrow T(A \Rightarrow TB)$. It follows $\vdash_{\mathsf{L}} (TA \Rightarrow TB) \Rightarrow T(A \Rightarrow TB)$.

■

Item 6 ($\Leftarrow$): Using (bind), we get $\vdash_{\mathsf{L}} (A \Rightarrow T(A \vee B)) \Rightarrow (TA \Rightarrow T(A \vee B))$ and $\vdash_{\mathsf{L}} (B \Rightarrow T(A \vee B)) \Rightarrow (TB \Rightarrow T(A \vee B))$. Using (unit), we derive $\vdash_{\mathsf{L}} TA \Rightarrow T(A \vee B)$ and $\vdash_{\mathsf{L}} TB \Rightarrow T(A \vee B)$. It follows $\vdash_{\mathsf{L}} (TA \vee TB) \Rightarrow T(A \vee B)$. We conclude using (bind).

■

Item 7 ($\Rightarrow$): We easily show $\vdash_{\mathsf{L}} (\forall x.TA) \Rightarrow TA[x \leftarrow y]$ for a fresh variable y. Using (bind) we get $\vdash_{\mathsf{L}} T(\forall x.TA) \Rightarrow TA[x \leftarrow y]$. We derive $T(\forall x.TA) \vdash_{\mathsf{L}} TA[x \leftarrow y]$ and therefore $T(\forall x.TA) \vdash_{\mathsf{L}} \forall x.TA$. We conclude $\vdash_{\mathsf{L}} T(\forall x.TA) \Rightarrow \forall x.TA$.

◀

3 Monad Translation

The standard double-negation translations [13, 12, 10, 14] have been defined for first-order logic. Several translations generalize them to monad operators [1, 6, 22] and extend them to higher-order logic [3, 21]. We want to define a translation that does both.

Kolmogorov's translation [13] and the Gödel-Gentzen translation [12, 10] cannot be extended to higher-order logic directly [3], because they do not preserve β-conversion. For example, let us take the Gödel-Gentzen translation of $(\lambda R.\ R \wedge Q)\ P \hookrightarrow P \wedge Q$. The translation of $(\lambda R.\ R \wedge Q)\ P$ is $(\lambda R.\ \neg\neg R \wedge \neg\neg Q)\ \neg\neg P$, which β-reduces to $\neg\neg\neg\neg P \wedge \neg\neg Q$, while the translation of $P \wedge Q$ is $\neg\neg P \wedge \neg\neg Q$. Unlike Kolmogorov's

translation and the Gödel-Gentzen translation, Kuroda's translation [14] preserves β-conversion and can be directly extended to higher-order logic [3, 21].

Moreover, a Kuroda-style monad translation [22] has been defined for first-order logic. It corresponds to Kuroda's translation, in which we insert monad operators instead of double negations. It also introduces additional monad operators after implications, and such modification is necessary to generalize Kuroda's translation to any monad operator. We will see in Section 7 how to remove this constraint for particular monad operators.

We define here a Kuroda-style monad translation for higher-order logic, taking advantage of the ideas developed in [21] and [22].

Definition 3 (Monad translation for higher-order logic).
Let A be a formula in higher-order logic and T be a monad operator. Its monad translation is $A^T := TA_T$, where $t \mapsto t_T$ is inductively defined by:

$$\begin{array}{lcl} x_T & := & x \\ c_T & := & \begin{cases} \lambda p.\ \forall x. T(p\ x) & \text{if } c = \forall \\ \lambda p.\ \lambda q.\ p \Rightarrow Tq & \text{if } c = \Rightarrow \\ c & \text{otherwise} \end{cases} \\ (\lambda x.\ t)_T & := & \lambda x.\ t_T \\ (t\ u)_T & := & t_T\ u_T \end{array}$$

In first-order logic, we have $(A[z \leftarrow w])^T = A^T[z \leftarrow w]$. As we are in higher-order logic, the substituted term w may be modified by the translation. We would intuitively expect $(A[z \leftarrow w])^T = A^T[z \leftarrow w^T]$, but we actually have $(A[z \leftarrow w])^T = A^T[z \leftarrow w_T]$, because the monad operator in front of the formula is inserted after the inductive translation. It follows that the translation $t \mapsto t_T$ commutes with substitution, but not $A \mapsto A^T$. Both translations $t \mapsto t_T$ and $A \mapsto A^T$ preserve convertibility.

Proposition 4.
For any term t, we have $(t[z \leftarrow w])_T = t_T[z \leftarrow w_T]$.

Proof.
By induction on the term t. ◀

Corollary 5.
For any higher-order formula A, we have $(A[z \leftarrow w])^T = A^T[z \leftarrow w_T]$.

In higher-order logic, terms are considered modulo β-convertibility. It follows that the translation must preserve β-convertibility for it to preserve provability.

Proposition 6.
For any terms t and u, if $t \equiv_\beta u$ then $t_T \equiv_\beta u_T$.

Proof.
By definition $((\lambda x.\ t)\ u)_T = (\lambda x.\ t_T)\ u_T$, and we have $(\lambda x.\ t_T)\ u_T \hookrightarrow t_T[x \leftarrow u_T]$. Using Proposition 4, we get $((\lambda x.\ t)\ u)_T \equiv_\beta (t[x \leftarrow u])_T$. Closure by context, reflexivity, symmetry, and transitivity are immediate. ◀

Corollary 7.
For any higher-order formulas A and B, if $A \equiv_\beta B$ then $A^T \equiv_\beta B^T$.

4 Monad Embedding

The monad translation generalizes double-negation translations with monad operators instead of double negations. Likewise, the monad translation eliminates the use of a particular formula [1, 6], called T-elimination, while double-negation translations eliminate the use of the principle of excluded middle.

$TA \Rightarrow A$ (T-elimination)

More formally, we extend the logic L with the inference rule T-Elim, and we write $\Gamma \vdash_{\mathsf{L}+\mathsf{T}} A$ when $\Gamma \vdash A$ is derivable using the rules of L and T-Elim.

$$\frac{}{\Gamma \vdash TA \Rightarrow A}\ \text{T-Elim}$$

The monad translation eliminates any use of the T-Elim inference rule–this is the soundness property. The characterization property states that any formula and its translation are equivalent assuming T-Elim.

When extending the double-negation elimination to higher-order logic [21], the characterization property does not trivially hold, but does hold when we consider functional extensionality and propositional extensionality. The same reasoning applies for the extension of the monad translation to higher-order logic.

Theorem 8 (Monad embedding).
Let T be a monad operator of L such that we have $\vdash_{\mathsf{L}} (TA)_T \Leftrightarrow TA_T$ for any formula A. Let Γ, A be higher-order formulas.
1.
If $\Gamma \vdash_{\mathsf{L}+\mathsf{T}} A$ then $\Gamma_T \vdash_{\mathsf{L}} A^T$.
2.
$\vdash^{\mathfrak{efp}}_{\mathsf{L}+\mathsf{T}} A^T \Leftrightarrow A$.

► **Remark 9.**
For Kolmogorov's translation and the Gödel-Gentzen translation, the soundness property is stated with Γ^T instead of Γ_T. When considering Kuroda's translation, we can actually omit the monad operators at the head of the formulas of the context. The version with Γ^T is provable as well.

Proof of Theorem 8(1).
We proceed by induction on the derivation. Most of the cases are direct applications of Proposition 2. Conv derives from Corollary 7. We only show the most interesting cases:
■
Or-E: Suppose that $\Gamma_T \vdash_{\mathsf{L}} T(A_T \vee B_T)$ and $\Gamma_T, A_T \vdash_{\mathsf{L}} TC_T$ and $\Gamma_T, B_T \vdash_{\mathsf{L}} TC_T$. To prove $\Gamma_T \vdash_{\mathsf{L}} TC_T$, it suffices to have $\Gamma_T \vdash_{\mathsf{L}} T(A_T \vee B_T) \Rightarrow TC_T$.
Using the second and third hypotheses, we derive $\Gamma_T \vdash_{\mathsf{L}} (A_T \vee B_T) \Rightarrow TC_T$. Using (bind) we get $\Gamma_T \vdash_{\mathsf{L}} T(A_T \vee B_T) \Rightarrow TC_T$ and we conclude the case.
■
All-I: Suppose $\Gamma_T \vdash_{\mathsf{L}} TA_T$. We derive $\Gamma_T \vdash_{\mathsf{L}} \forall x.TA^T$, that is $\Gamma_T \vdash_{\mathsf{L}} (\forall x.A)_T$. We conclude using (unit).
■
All-E: Suppose $\Gamma_T \vdash_{\mathsf{L}} T(\forall x.TA_T)$. Using Proposition 2(7), we get $\Gamma_T \vdash_{\mathsf{L}} \forall x.TA_T$. From All-E we obtain $\Gamma_T \vdash_{\mathsf{L}} (TA_T)[x \leftarrow t_T]$. We conclude with Corollary 5.
■
Ex-I: Suppose $\Gamma_T \vdash_{\mathsf{L}} T(A[x \leftarrow t])_T$. Using Corollary 5, we get $\Gamma_T \vdash_{\mathsf{L}} TA_T[x \leftarrow t_T]$. From Ex-I we obtain $\Gamma_T \vdash_{\mathsf{L}} \exists x.TA_T$. We conclude using (unit) and Proposition 2(8).
■
Ex-E: Suppose $\Gamma_T \vdash_{\mathsf{L}} T(\exists x.A_T)$ and $\Gamma_T, A_T \vdash_{\mathsf{L}} TC_T$. To conclude the proof, it suffices to have $\Gamma_T \vdash_{\mathsf{L}} T(\exists x.A_T) \Rightarrow TC_T$.
We have $\Gamma_T, \exists x.A_T \vdash_{\mathsf{L}} \exists x.A_T$ (by Ax) and $\Gamma_T, \exists x.A_T, A_T \vdash_{\mathsf{L}} TC_T$ (by weakening of the second hypothesis). Using Ex-E, we derive $\Gamma_T, \exists x.A_T \vdash_{\mathsf{L}} TC_T$ and therefore $\Gamma_T \vdash_{\mathsf{L}} (\exists x.A_T) \Rightarrow TC_T$. Using (bind), we obtain $\Gamma_T \vdash_{\mathsf{L}} T(\exists x.A_T) \Rightarrow TC_T$ and we conclude the case.
■
Bot-E: Suppose $\Gamma_T \vdash_{\mathsf{IL}} T\bot$. As we are in intuitionistic logic, we have $\Gamma_T \vdash_{\mathsf{IL}} \bot \Rightarrow A_T$. We deduce $\Gamma_T \vdash_{\mathsf{IL}} T\bot \Rightarrow TA_T$ using Proposition 2(2). We conclude $\Gamma_T \vdash_{\mathsf{IL}} TA_T$.
■
PEM: As we are in classical logic, we have $\Gamma_T \vdash_{\mathsf{CL}} A_T \vee (A_T \Rightarrow \bot)$. Using (unit), we derive $\Gamma_T \vdash_{\mathsf{CL}} A_T \vee (A_T \Rightarrow T\bot)$ and then we conclude $\Gamma_T \vdash_{\mathsf{CL}} T(A_T \vee (A_T \Rightarrow T\bot))$.
■
T-Elim: We have $(TA \Rightarrow A)_T = ((TA)_T \Rightarrow TA_T)$. We know $\vdash_{\mathsf{L}} (TA)_T \Leftrightarrow TA_T$, so we derive $\Gamma_T \vdash_{\mathsf{L}} (TA \Rightarrow A)_T$. We conclude $\Gamma_T \vdash_{\mathsf{L}} T(TA \Rightarrow A)_T$ using (unit).
◄

Before proving the second item of Theorem 8, we show the following lemma, stating that any term and its translation are equal, assuming T-Elim, functional extensionality and propositional extensionality.

Lemma 10.
For any term t, we have $\vdash^{\mathfrak{efp}}_{\mathsf{L}+\mathsf{T}} t_T = t$.

Proof.
We proceed by induction on the term t. We directly have $\vdash^{\mathfrak{efp}}_{\mathsf{L}+\mathsf{T}} x_T = x$ and $\vdash^{\mathfrak{efp}}_{\mathsf{L}+\mathsf{T}} c_T = c$ for $c \notin \{\forall, \Rightarrow\}$. For $c \in \{\forall, \Rightarrow\}$, we derive $\vdash^{\mathfrak{efp}}_{\mathsf{L}+\mathsf{T}} c_T = c$ from PropExt, FunExt, T-Elim and (unit). We have

$\vdash^{\mathfrak{efp}}_{\mathsf{L}+\mathsf{T}} (t\ u)_T = t\ u$ using the induction hypotheses and Eq-E. We derive $\vdash^{\mathfrak{efp}}_{\mathsf{L}+\mathsf{T}} (\lambda x.\ t)_T = \lambda x.\ t$ using the induction hypothesis and FunExt. ◀

Proof of Theorem 8(2).

We derive $\vdash^{\mathfrak{efp}}_{\mathsf{L}+\mathsf{T}} A_T \Leftrightarrow A$ using Lemma 10, and then we conclude $\vdash^{\mathfrak{efp}}_{\mathsf{L}+\mathsf{T}} A^T \Leftrightarrow A$ using (unit) and T-Elim. ◀

We proved that the monad translation satisfies the soundness and characterization properties. For the characterization property, we had to introduce symbols of equality and to assume both functional extensionality and propositional extensionality. But does the soundness property still hold in the presence of equality?

The proof of the soundness property does not naturally extend when we consider the symbols and inference rules of equality. Let us consider the formulas

$\forall x \forall y.\ T(x = y) \Rightarrow x = y \qquad (\Delta^{\mathfrak{ef}})$

$\forall x \forall y.\ (Tx = Ty) \Rightarrow x = y \qquad (\Delta^{\mathfrak{ep}})$

regarding T and equality predicates. We write $\Delta^{\mathfrak{efp}}$ for the context $\Delta^{\mathfrak{ef}}, \Delta^{\mathfrak{ep}}$. When proving the soundness property in the presence of equality, the first formula $\Delta^{\mathfrak{ef}}$ is useful for the FunExt case and the second formula $\Delta^{\mathfrak{ep}}$ is useful for the PropExt case.

Theorem 11 (Soundness for equality).

Let T be a monad operator of L such that we have $\vdash_{\mathsf{L}} (TA)_T \Leftrightarrow TA_T$ for any formula A. Let Γ, A be higher-order formulas.

1.

If $\Gamma \vdash^{\mathfrak{e}}_{\mathsf{L}+\mathsf{T}} A$ then $\Gamma_T \vdash^{\mathfrak{e}}_{\mathsf{L}} A^T$.

2.

For $* \in \{\mathfrak{ef}, \mathfrak{ep}, \mathfrak{efp}\}$, if $\Gamma \vdash^{*}_{\mathsf{L}+\mathsf{T}} A$ then $\Delta^*, \Gamma_T \vdash^{*}_{\mathsf{L}} A^T$.

Proof.

For the first item, we complete the proof of Theorem 8 with the following cases:

■

Eq-I: We directly use Eq-I and (unit).

■

Eq-E: Suppose $\Gamma_T \vdash^*_{\mathsf{L}} TA_T[x \leftarrow u_T]$ and $\Gamma_T \vdash^*_{\mathsf{L}} T(u_T = v_T)$. Using Eq-E, we derive $\vdash^*_{\mathsf{L}} A_T[x \leftarrow u_T] \Rightarrow (u_T = v_T) \Rightarrow A_T[x \leftarrow v_T]$. Therefore, we get $\vdash^*_{\mathsf{L}} TA_T[x \leftarrow u_T] \Rightarrow T(u_T = v_T) \Rightarrow TA_T[x \leftarrow v_T]$ using Proposition 2(2) and Proposition 2(4). We conclude $\Gamma_T \vdash^*_{\mathsf{L}} TA_T[x \leftarrow v_T]$.

For the second item, most cases are those of the first item that are reused using weakening on Δ^*. The remaining cases are:

■

FunExt: Suppose $\Delta^*, \Gamma_T \vdash^*_{\mathsf{L}} T(f_T\ x = g_T\ x)$. Since $x \notin \mathrm{FV}(\Gamma, f, g)$, we directly have $x \notin \mathrm{FV}(\Gamma_T, f_T, g_T)$. We get $\Delta^*, \Gamma_T \vdash^*_{\mathsf{L}} f_T\ x = g_T\ x$ using $\Delta^{\mathfrak{ef}}$. We conclude using FunExt and (unit).

■

PropExt: Suppose $\Delta^*, \Gamma_T \vdash^*_{\mathsf{L}} T(A_T \Rightarrow TB_T)$ and $\Delta^*, \Gamma_T \vdash^*_{\mathsf{L}} T(B_T \Rightarrow TA_T)$. Using Proposition 2(5), we get $\Delta^*, \Gamma_T \vdash^*_{\mathsf{L}} TA_T \Rightarrow TB_T$ and $\Delta^*, \Gamma_T \vdash^*_{\mathsf{L}} TB_T \Rightarrow TA_T$. Using PropExt we derive $\Delta^*, \Gamma_T \vdash^*_{\mathsf{L}} TA_T = TB_T$. We conclude using $\Delta^{\mathfrak{ep}}$ and (unit).

◀

When extending Kuroda's double-negation translation to higher-order logic [21], we only considered the formula $\Delta^{\mathfrak{ef}}$. The generalization to any monad operator requires us to additionally consider the formula $\Delta^{\mathfrak{ep}}$. In Section 7, we will see that this additional assumption can be dropped when considering factorizable monad operators.

5 Embeddings between Logics

Such monad translation can be directly applied with particular monad operators to embed classical logic into intuitionistic logic or intuitionistic logic into minimal logic.

There are many classical laws, such as the principle of excluded middle, the double-negation elimination, Peirce's law, or Clavius's law. All these laws are equivalent in intuitionistic logic.

$\neg\neg A \Rightarrow A$ (double-negation elimination)

$(\neg A \Rightarrow A) \Rightarrow A$ (Clavius's law)

$((A \Rightarrow B) \Rightarrow A) \Rightarrow A$ (Peirce's law)

The T-elimination principle corresponds to the double-negation elimination when we choose $TA := (A \Rightarrow \bot) \Rightarrow \bot$ and to Clavius's law when we choose $TA := (A \Rightarrow \bot) \Rightarrow A$. For these monad operators, the T-Elim inference rule is therefore equivalent to PEM, and eliminating any use of T-elim in the derivations

means transforming classical provability into intuitionistic provability. When $T_R A := (A \Rightarrow R) \Rightarrow A$, the T-elimination principle only corresponds to an instance of Peirce's law with $B := R$.

Corollary 12 (Embeddings of classical logic into intuitionistic logic).
Let Γ, A be higher-order formulas.

1. If $\Gamma \vdash_{\mathsf{CL}} A$ then $\Gamma_T \vdash_{\mathsf{IL}} A^T$ with $TA := (A \Rightarrow \bot) \Rightarrow \bot$.
2. If $\Gamma \vdash_{\mathsf{CL}} A$ then $\Gamma_T \vdash_{\mathsf{IL}} A^T$ with $TA := (A \Rightarrow \bot) \Rightarrow A$.
3. If $\Gamma \vdash_{\mathsf{CL}} A$ then there exists a formula R such that $\Gamma_{T_R} \vdash_{\mathsf{IL}} A^{T_R}$ with $T_R A := (A \Rightarrow R) \Rightarrow A$.

Moreover, for each of the above, we have $\vdash^{\mathfrak{efp}}_{\mathsf{CL}} A^T \Leftrightarrow A$.

Proof.
We apply Theorem 8 with $\mathsf{L} := \mathsf{IL}$. T-Elim is equivalent to PEM, so $\mathsf{IL} + \mathsf{T}$ corresponds to CL. For the third item, we have to give the correct parameter R. The derivation $\Gamma \vdash_{\mathsf{CL}} A$ is finite, so Peirce's law is used a finite number of times with formulas $B_1, \dots, B_n$ of free variables $\vec{x_1}, \dots, \vec{x_n}$. Such free variables could be captured later in the derivation. As remarked by Escardó and Oliva [6], $T_{R_1 \wedge R_2} A \Rightarrow A \vdash_{\mathsf{IL}} (T_{R_1} A \Rightarrow A) \wedge (T_{R_2} A \Rightarrow A)$. With the additional remark that $T_{\forall x.R} A \Rightarrow A \vdash_{\mathsf{IL}} T_{R[x \leftarrow t]} A \Rightarrow A$, we define R to be the conjunction $\forall \vec{x_1}.B_1 \wedge \ldots \wedge \forall \vec{x_n}.B_n$, so that the monad translation eliminates the n instances of Peirce's law that occur in the derivation. ◀

When $TA := A \vee \bot$, the T-elimination principle is equivalent to the principle of explosion, and the T-Elim inference rule is therefore equivalent to the Bot-E inference rule. In that case, eliminating any use of T-elim in the derivations means transforming intuitionistic provability into minimal provability.

Corollary 13 (Embedding of intuitionistic logic into minimal logic).
Let Γ, A be higher-order formulas. If $\Gamma \vdash_{\mathsf{IL}} A$ then $\Gamma_T \vdash_{\mathsf{ML}} A^T$ with $TA := A \vee \bot$. Moreover, we have $\vdash^{\mathfrak{efp}}_{\mathsf{IL}} A^T \Leftrightarrow A$.

Proof.
We apply Theorem 8 with $\mathsf{L} := \mathsf{ML}$. T-Elim is equivalent to Bot-E, so $\mathsf{ML} + \mathsf{T}$ corresponds to IL. ◀

6 Fragment of Coherent Formulas

Coherent formulas are formulas that can only be built using conjunctions, disjunctions and existential quantifiers. Because monad operators are only inserted after universal quantifiers and implications, it follows that $A_T = A$ for any coherent formula A. We lift to higher-order logic two tricks for coherent formulas that rely on this observation.

The first trick [5, 22] allows us to derive minimal provability from intuitionistic provability when the context is empty and when the formula is coherent.

Corollary 14.
Let Γ, A be higher-order coherent formulas. If $\vdash_{\mathsf{IL}} A$ then $\vdash_{\mathsf{ML}} A$.

Proof.
Suppose $\vdash_{\mathsf{IL}} A$. We apply Theorem 8 with $TA := A \vee \bot$, and we derive $\vdash_{\mathsf{ML}} TA_T$, that is $\vdash_{\mathsf{ML}} A \vee \bot$. We know that minimal logic has the disjunction property, and that we cannot prove $\bot$. Hence, we conclude $\vdash_{\mathsf{ML}} A$. ◀

The second trick [18, 22] allows us to derive intuitionistic provability from classical provability when the formulas involved are coherent. It corresponds to a weaker version–with finite disjunctions–of Barr's theorem [2] extended to higher-order logic.

Theorem 15.
Let Γ, A be higher-order coherent formulas. If $\Gamma \vdash_{\mathsf{CL}} A$ then $\Gamma \vdash_{\mathsf{IL}} A$.

To prove this result, we adapt to higher-order logic Palmgren's idea [18] of using Friedman's translation [9]. As an intermediary lemma, we prove that, for higher-order logic, Friedman's translation transforms classical proofs into intuitionistic proofs. Remark that we cannot directly apply Theorem 8, as the T-elimination principle does not correspond to a classical law when $TA := (A \Rightarrow R) \Rightarrow R$.

Lemma 16.
Let $TA := (A \Rightarrow R) \Rightarrow R$ with parameter R. Let Γ, A be higher-order formulas. If $\Gamma \vdash_{\mathsf{CL}} A$ then $\Gamma_T \vdash_{\mathsf{IL}} A^T$.

Proof.

We proceed by induction on the derivation. All the cases are similar to the cases of Theorem 8, except that we do not need the T-Elim case anymore, and that we change the PEM case. We prove $\Gamma_T \vdash_{\mathsf{IL}} T(A \vee \neg A)_T$ using

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma_T, B \Rightarrow R, A_T, \bot \Rightarrow R \vdash_{\mathsf{IL}} A_T}\,\textsc{Ax}}{\Gamma_T, B \Rightarrow R, A_T, \bot \Rightarrow R \vdash_{\mathsf{IL}} B}\,\textsc{Or-IL}}{\Gamma_T, B \Rightarrow R, A_T, \bot \Rightarrow R \vdash_{\mathsf{IL}} R}\,\textsc{Imp-E with } B \Rightarrow R}{\Gamma_T, B \Rightarrow R \vdash_{\mathsf{IL}} A_T \Rightarrow ((\bot \Rightarrow R) \Rightarrow R)}\,\textsc{Imp-I} \times 2}{\Gamma_T, B \Rightarrow R \vdash_{\mathsf{IL}} B}\,\textsc{Or-IR}}{\Gamma_T, B \Rightarrow R \vdash_{\mathsf{IL}} R}\,\textsc{Imp-E with } B \Rightarrow R}{\Gamma_T \vdash_{\mathsf{IL}} ((A \vee (A \Rightarrow ((\bot \Rightarrow R) \Rightarrow R))) \Rightarrow R) \Rightarrow R}\,\textsc{Imp-I}$$

where B is an abbreviation for the formula $A_T \vee (A_T \Rightarrow ((\bot \Rightarrow R) \Rightarrow R))$. ◀

Proof of Theorem 15.

Suppose $\Gamma \vdash_{\mathsf{CL}} A$. Without loss of generality, we assume that Γ and A have no free variables, as we can replace them by fresh constants. We apply Lemma 16, and we derive $\Gamma_{T_R} \vdash_{\mathsf{IL}} T_R A_{T_R}$, that is $\Gamma \vdash_{\mathsf{IL}} (A \Rightarrow R) \Rightarrow R$. Choosing $R := A$, we get $\Gamma \vdash_{\mathsf{IL}} A$. ◀

In other words, coherent formulas correspond to a constructive fragment of higher-order classical logic. Constructive fragments of classical logic have been studied for propositional logic [11, 12] and first-order logic [19]. This is the first time, to our knowledge, that such results are proved for higher-order logic.

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